

Complex Systems Excercise 1

Idan Stark

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1 Dimensional analysis - Flow in porous media

Soap films are fluid membranes, whose shape in space is dominated by surface tension. As such, an unconstrained soap film (not enclosing any volume) will form a minimal surface.

Let us consider two identical rings of radius R dipped together in soap-water, and then drawn apart from each other. Assuming they make an open surface, and assuming the symmetry of the problem is maintained, let us look for a solution in the form of a surface of revolution:

$$\vec{r}(z, \nu) = \phi(z) \cos(\nu) \hat{x} + \phi(z) \sin(\nu) \hat{y} + z \hat{z} \quad (1)$$

where $\nu \in [0, 2\pi]$ and $\phi(z)$ is the profile function.

1.1 Tangent vectors

The tangent vectors of the surface are given by:

$$\begin{aligned} \vec{r}_z &= \partial_z \vec{r} = \phi'(z) \cos(\nu) \hat{x} + \phi'(z) \sin(\nu) \hat{y} + \hat{z} \\ \vec{r}_\nu &= \partial_\nu \vec{r} = -\phi(z) \sin(\nu) \hat{x} + \phi(z) \cos(\nu) \hat{y} \end{aligned} \quad (2)$$

1.2 The metric tensor

This gives:

$$\begin{aligned} a_{zz} &= \vec{r}_z \cdot \vec{r}_z = \phi'^2(z) \cos^2(\nu) + \phi'^2(z) \sin^2(\nu) + 1^2 = 1 + \phi'^2(z) \\ a_{\nu z} &= a_{z\nu} = \vec{r}_\nu \cdot \vec{r}_z = -\phi(z) \phi'(z) \cos(\nu) \sin(\nu) + \phi(z) \phi'(z) \cos(\nu) \sin(\nu) + 0 = 0 \\ a_{\nu\nu} &= \vec{r}_\nu \cdot \vec{r}_\nu = \phi^2(z) \sin^2(\nu) + \phi^2(z) \cos^2(\nu) = \phi^2(z) \end{aligned} \quad (3)$$

So we get:

$$a_{\alpha\beta} = \begin{pmatrix} 1 + \phi'^2 & \\ & \phi^2 \end{pmatrix} \quad (4)$$

1.3 The bending tensor and shape tensor

First, let us find the normal vector:

$$\hat{N} = \frac{\vec{r}_z \times \vec{r}_\nu}{\sqrt{a}} = \frac{-\phi(z) \cos(\nu) \hat{x} - \phi(z) \sin(\nu) \hat{y} + \phi(z) \phi'(z) \hat{z}}{\sqrt{(1 + \phi'^2) \phi^2}} \quad (5)$$

Also:

$$\begin{aligned} \partial_z \partial_z \vec{r} &= \phi''(z) \cos(\nu) \hat{x} + \phi''(z) \sin(\nu) \hat{y} \\ \partial_z \partial_\nu \vec{r} &= \partial_\nu \partial_z \vec{r} = -\phi'(z) \sin(\nu) \hat{x} + \phi'(z) \cos(\nu) \hat{y} \\ \partial_\nu \partial_\nu \vec{r} &= -\phi(z) \cos(\nu) \hat{x} - \phi(z) \sin(\nu) \hat{y} \end{aligned} \quad (6)$$

Which gives:

$$b_{\alpha\beta} = \partial_\alpha \partial_\beta \vec{r} \cdot \hat{N} = \frac{1}{\sqrt{1 + \phi'^2}} \begin{pmatrix} -\phi'' & \\ & \phi \end{pmatrix} \quad (7)$$

And the shape tensor is:

$$s_\beta^\alpha = \frac{1}{\sqrt{1 + \phi'^2}} \begin{pmatrix} \frac{1}{1 + \phi'^2} & \\ & \frac{1}{\phi^2} \end{pmatrix} \begin{pmatrix} -\phi'' & \\ & \phi \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{1 + \phi'^2}} \begin{pmatrix} -\frac{\phi''}{1 + \phi'^2} & \\ & \frac{1}{\phi} \end{pmatrix} \end{pmatrix} \quad (8)$$

1.4 Minimal surface

For a minimal (open) surface, we require

$$H = 0 \rightarrow \text{Tr}\{s\} = 0 \quad (9)$$

This gives the equation:

$$-\frac{\phi''}{1 + \phi'^2} + \frac{1}{\phi} = 0 \quad (10)$$

1.5 Solving the ODE

Multiplying and organizing gives:

$$\phi \phi'' - \phi'^2 - 1 = 0 \quad (11)$$

Using:

$$\phi'' = \frac{d\phi'}{dx} = \frac{d\phi'}{d\phi} \frac{d\phi}{dx} = \phi' \frac{d\phi'}{d\phi} = \frac{1}{2} \frac{d}{d\phi} \phi'^2 \quad (12)$$

We can define the derivative as a function of ϕ to get $\sigma(\phi) = \phi'^2(\phi)$. The ODE then reduces to a linear ODE:

$$\phi \frac{d\sigma}{d\phi} - 2\sigma - 2 = 0 \quad (13)$$

Separating:

$$\frac{d\sigma}{2\sigma + 2} = \frac{d\phi}{\phi} \quad (14)$$

Integrating:

$$\frac{1}{2} \ln(\sigma + 1) = \ln \phi + C \quad (15)$$

And organizing:

$$\sigma = A\phi^2 - 1 \quad (16)$$

Then plugging back the definition:

$$\phi' = \sqrt{A\phi^2 - 1} \quad (17)$$

This is simpler equation, but to make it even easier let us find A using the boundary conditions on $z = 0$. This will give $\phi' = 0, \phi = r_{min}$ and thus:

$$Ar_{min}^2 - 1 = 0 \rightarrow A = r_{min}^{-2} \quad (18)$$

So, we can define $\varphi = \phi/r_{min}$ to get:

$$r_{min}\varphi' = \sqrt{\varphi^2 - 1} \quad (19)$$

Using separation of variables again:

$$r_{min} \frac{d\varphi}{\sqrt{\varphi^2 - 1}} = dz \quad (20)$$

And integrating

$$r_{min} \cosh^{-1} \varphi = z + B \quad (21)$$

And finally solving for ϕ gives:

$$\phi = r_{min} \cosh\left(\frac{z + B}{r_{min}}\right) \quad (22)$$

Plugging $\phi(0) = r_{min}$ gives $B = 0$, and so we get:

$$\boxed{\phi = r_{min} \cosh\left(\frac{z}{r_{min}}\right)} \quad (23)$$

1.6 Resulting metric and shape tensors

We can plug this result into the metric and shape tensors, and use $1 - \sinh^2(x) = \cosh^2(x)$ to get:

$$\begin{aligned} a_{\alpha\beta} &= \begin{pmatrix} 1 + \sinh^2(z/r_{min}) & \\ & r_{min}^2 \cosh^2(z/r_{min}) \end{pmatrix} = \\ &= \cosh^2(z/r_{min}) \begin{pmatrix} 1 & \\ & r_{min}^2 \end{pmatrix} \end{aligned} \quad (24)$$

$$\begin{aligned} s_{\beta}^{\alpha} &= \frac{1}{\sqrt{1 + \sinh^2(z/r_{min})}} \begin{pmatrix} -\frac{\cosh(z/r_{min})}{r_{min}(1 + \sinh^2(z/r_{min}))} & \\ & \frac{1}{r_{min} \cosh(z/r_{min})} \end{pmatrix} = \\ &= \frac{1}{r_{min} \cosh^2(z/r_{min})} \begin{pmatrix} -1 & \\ & 1 \end{pmatrix} \end{aligned} \quad (25)$$

To summarize:

$$\boxed{a_{\alpha\beta} = \cosh^2(z/r_{min}) \begin{pmatrix} 1 & \\ & r_{min}^2 \end{pmatrix} \quad s_{\beta}^{\alpha} = \frac{1}{r_{min} \cosh^2(z/r_{min})} \begin{pmatrix} -1 & \\ & 1 \end{pmatrix}} \quad (26)$$

1.7 Given rings

If the two rings have a given radius R , and are placed at $z = \pm z_0$, we can plug this into the $\phi(z)$ to get:

$$\boxed{r_{min} \cosh(z_0/r_{min}) = R} \quad (27)$$

1.8 Explaining the video

In the video we see the shape that we calculated. The rings start with $z_0 \ll R$, and then as they grow apart and z_0 grows, to keep R constant, r_{min} must shrink.